

Day 36



JavaScript
THE LANGUAGE OF THE WEB

Cookies in JavaScript:

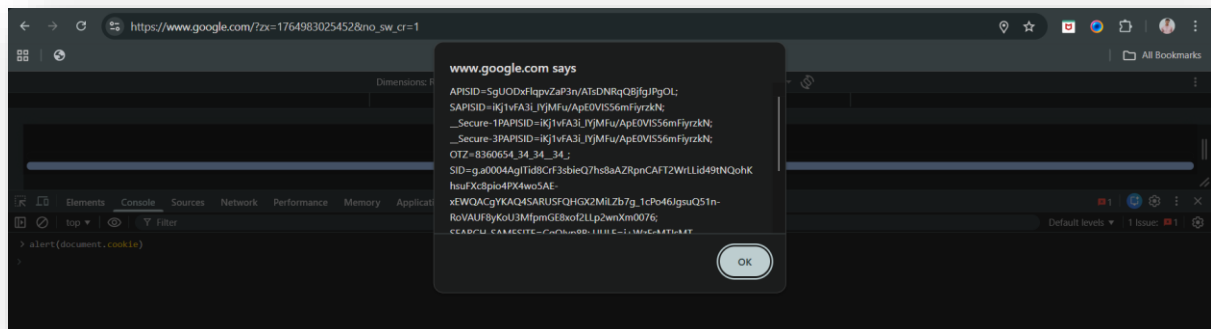
What are Cookies?

Cookies are small pieces of data stored in the browser. Websites use them to remember things like:

- Login information
- User preferences
- Items in a cart
- Last visited page

Cookies are stored as key–value pairs.

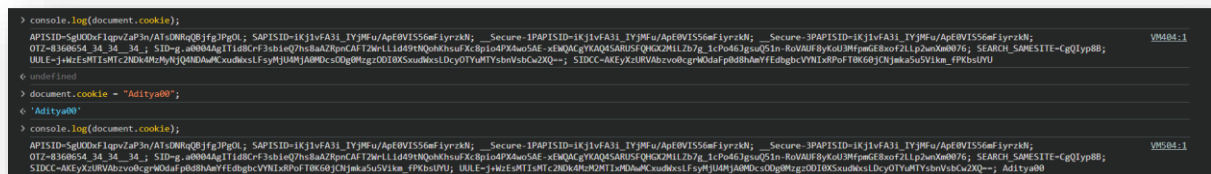
Where can we find cookie for a website?



How to Create/Set a Cookie

You set a cookie by writing to document.cookie. By doing so we are adding cookies to the existing cookies, not replacing any cookie.

Example:



Now, we can also take user input and set them as the part of the cookie:

```
> let key = prompt("Enter the key:");
< undefined
> let value = prompt("Enter the value:");
< undefined
> console.log(`${key}: ${value}`);
1_name: 9487193h1
< undefined
> console.log(document.cookie);
APISID=5qIXDnF1qpvZp3n/ATs0MRq0jfg2Pg0L; SAPISTID=1Kj1vFA31_IY9MFu/ApE0V1S56fIyrzK; _Secure-1PAPISID=1Kj1vFA31_IY9MFu/ApE0V1S56fIyrzK; _Secure-3PAPISID=1Kj1vFA31_IY9MFu/ApE0V1S56fIyrzK; V01426:1
OTZ-8368654_34_34_34; SID-g_a0004Ag1T1g8CrF3sb1eQ7h8aA2IpmCAF72M-L1G49H9QhKhsufXc8pioAPKAwSAE_xEMQACgYKAQ45ARUSFQhKX2M1L7b7g_1cPo46JguQ51n-RoVAUfByEol3MfpmGtBxofZLLp2wXm0076; SEARCH_SAME_SITE=CgQIyp88; Aditya00;
UULE-j+mzEsHTTsHTC2NDu4HzcyMTU0HzAwNCxudKcsLFsyMJU0HJA0MDCsODg0Hzgz0010XSxudKcsLDcy0TYuMTYsbWsbCw2XQ=-; SIDCC=AKExCt18KfaLpnj7j38SQJ28hgFNDKpabge0J1TIXkqp9rs1UPpycT0gY01tsKJXefd91v1vY
```

But, will it be able to take and store URI component and store by above method? No! We need to use encodeURIComponent to do so.

```
> let key = prompt("Enter the key:");
< undefined
> let value = prompt("Enter the value:");
< undefined
✖ Uncaught (in promise) Error: A listener indicated an asynchronous response by returning true, but the message cha
✖ Unchecked runtime.lastError: A listener indicated an asynchronous response by returning true, but the message cha
✖ Unchecked runtime.lastError: A listener indicated an asynchronous response by returning true, but the message cha
> console.log(`${encodeURIComponent(key)}: ${encodeURIComponent(value)}`);
name33: %26%25%3B
```

Also, to decode it we are using decodeURIComponent:

```
> decodeURIComponent('%26%25%3B')
< '&';'
```

Cookie Options:

Common attributes you can add:

Option	Meaning
expires	When cookie should expire
max-age	How long cookie stays (seconds)
path=	Cookie valid for whole site
secure	Only sent over HTTPS
SameSite	Controls cross-site cookie sending

We can't add multiple cookies just by separating them with “,”:

```
> document.cookie = "1=2;3=4";  
< '1=2;3=4'  
> console.log(document.cookie);  
APISID=5gU0xT1qovZAP3n/ATsDNRqBjfgJpGOL; SAPISID=IKJ1vFA31_IY3Mfu/ApE0V1556mfIyrzKH; __Secure-1PAPISID=IKJ1vFA31_IY3Mfu/ApE0V1556mfIyrzKH; __Secure-3PAPISID=IKJ1vFA31_IY3Mfu/ApE0V1556mfIyrzKH; W02168-1  
OTZ-4368654_34_34_34; SID=ga-8808Aag1I1u8C-F3ab1e-Q7nsb2ZgmCAFTZm-1L184918QmKnuFzK8q1uAPPAw0JME-AEQ0AqTYAQ4SARISFQK6ZMLL2D7g_1c-P0483gswQIn-RuNAUFRyKau3MfpmE2noFZ1Lp2anXo8R76; SEARCH_SAME SITE=CgQIyp8B; Aditya00; 1=2;  
URL E=j+MeEsmT1sMTc2NDU4NDMyNjUzZmTlAwPCxud8ocLFsyMjU0NjABMPC-cDg0RnzrODI0XSou8Boc1DcyOTY0MTYabnVsbCw2XQ==; STDC-AKEYXVWZ83YfEhugON07hg065V1xM89ac3LjqlGt3w81qrEIfcpc20u-k4PJE0J1cmKtQ
```

Important points about cookies:

- Cookies are automatically sent to the server with every request.
- Maximum size is about 4 KB per cookie.
- Cookies can expire using expires or max-age.
- Without expiry, cookies are deleted when the browser closes.
- Cookies are stored as plain text, so avoid storing sensitive data.
- Use secure, HttpOnly, and SameSite for better security.
- Updating a cookie is done by writing it again with the same name.
- Deleting a cookie requires setting an old expiration date.
- Cookies work across pages of the same domain (unless restricted).
- Cookies slow down requests because they are sent every time.
- Cookies store only strings, so objects must be converted using JSON.stringify().

--The End--